



WIRTSCHAFTS
UNIVERSITÄT
WIEN VIENNA
UNIVERSITY OF
ECONOMICS
AND BUSINESS

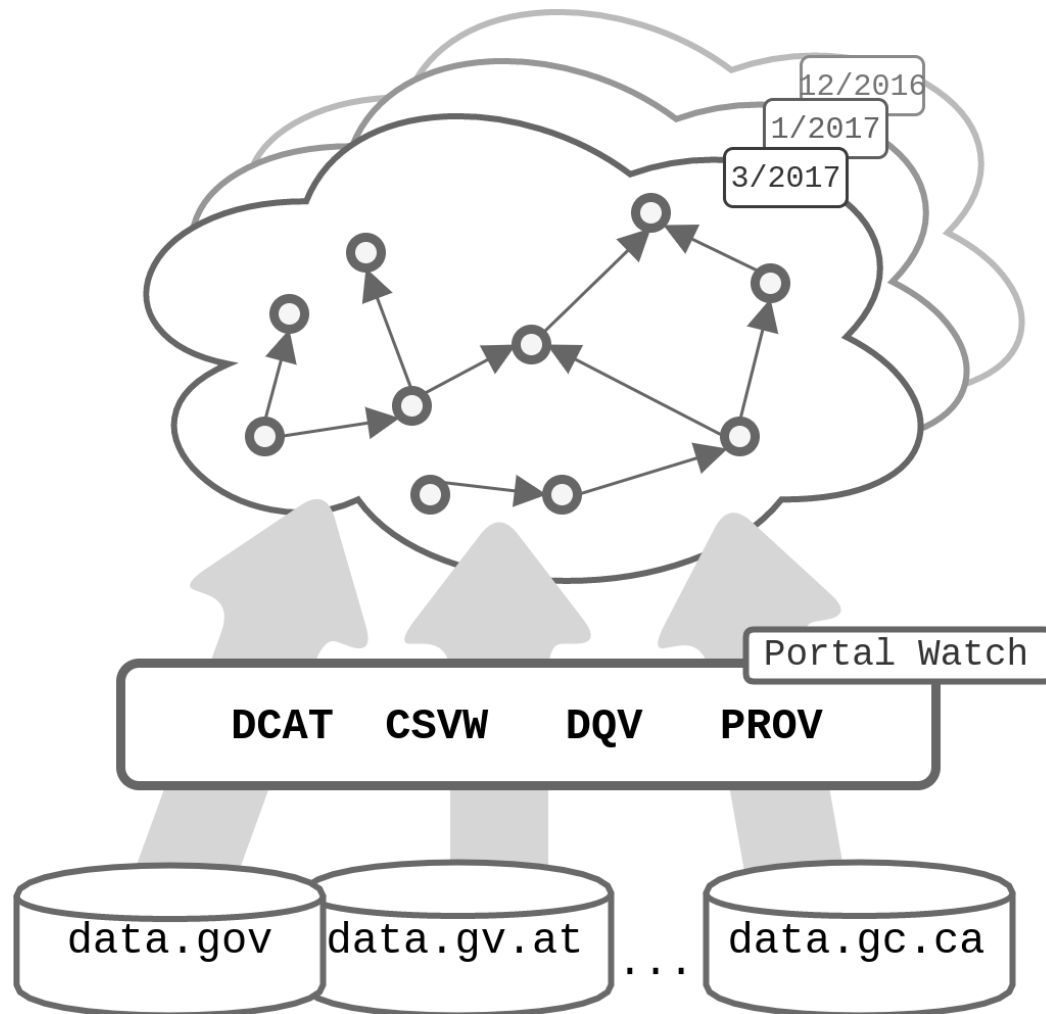


Lifting Data Portals to the Web of Data

Sebastian Neumaier, Jürgen Umbrich, Axel Polleres

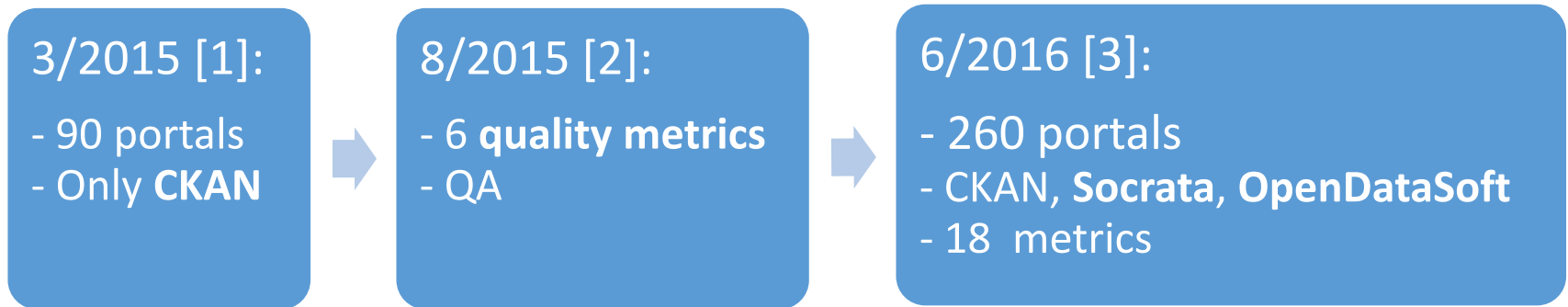
Vienna University of Economics and Business, Vienna, Austria

Lifting Data Portals to the Web of Data



Open Data Portal Watch Project

Monitoring and QA over evolving data portals



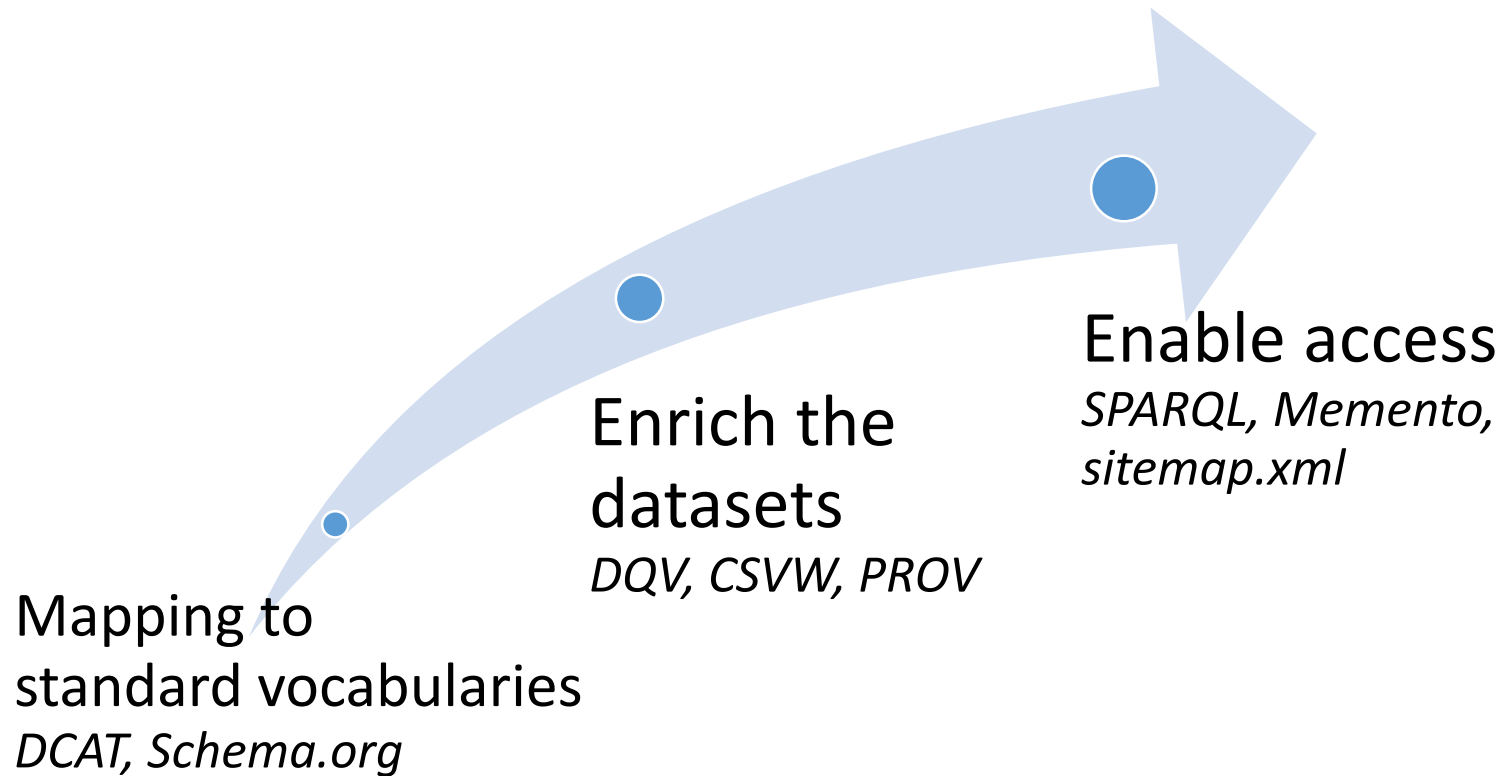
	total	CKAN	Socrata	ODSoft	DCAT
portals	261	149	99	11	2
datasets	854,013	767,364	81,268	3,340	2,041
URLs	2,057,924	1,964,971	104,298	12,398	6,092

- [1] Towards assessing the quality evolution of open data portals. In ODQ2015: Open Data Quality Workshop, Munich, Germany
- [2] Quality assessment & evolution of open data portals. In: International Conference on Open and Big Data, Rome, Italy (2015)
- [3] Automated quality assessment of metadata across open data portals. ACM Journal of Data and Information Quality (2016) ³

Identified challenges

- Metadata is **heterogeneous** and (partially) messy
 - Software-specific metadata (CKAN vs Socrata vs ...)
 - Portal-specific metadata
 - Missing metadata (file formats, API descriptions, ...)
- Metadata not available as Linked Data
 - Only partially in DCAT vocabulary
 - **No mappings** for additional metadata fields
- Poor **discoverability** of datasets
 - No content information in metadata (e.g., CSV headers)
 - Datasets' metadata not optimized for search engines

Current progress

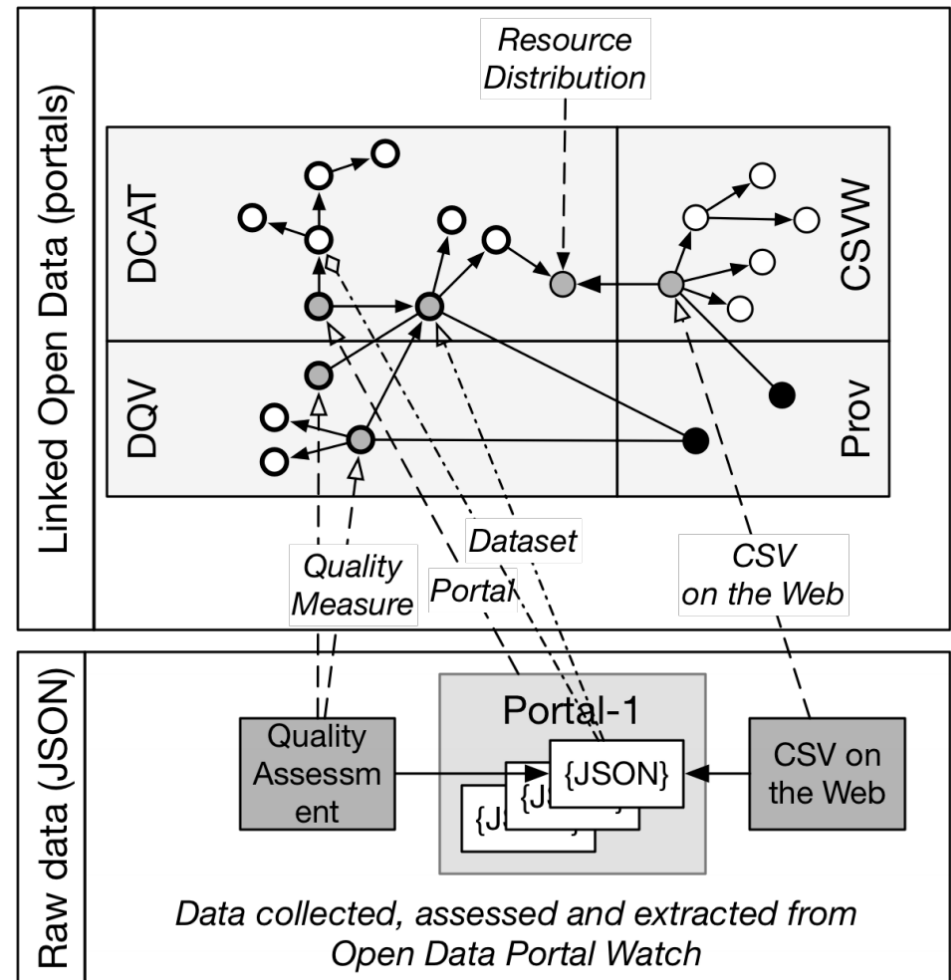


Mapping to standard vocabularies

- **DCAT** export of metadata from CKAN, Socrata and OpenDataSoft portals
 - Mapping of most frequent (portal/domain specific) extra-metadata fields
- Mapping and publishing of **Schema.org**:
 - Mapping of DCAT to Schema.org's dataset vocabulary
 - Enabling integration into knowledge graphs of major search engines

Enrich the datasets

- *Portal Watch quality dimensions:*
 - **Data Quality vocabulary**
- *Metadata for tables:*
 - **CSV on the Web vocabulary**
- *Record provenance:*
 - **PROV ontology**



CSV on the Web metadata

- Dialect properties:

HTTP Content-Type

→ dcat:mediaType

Encoding detection

→ csvw:encoding

Delimiter detection

→ csvw:delimiter

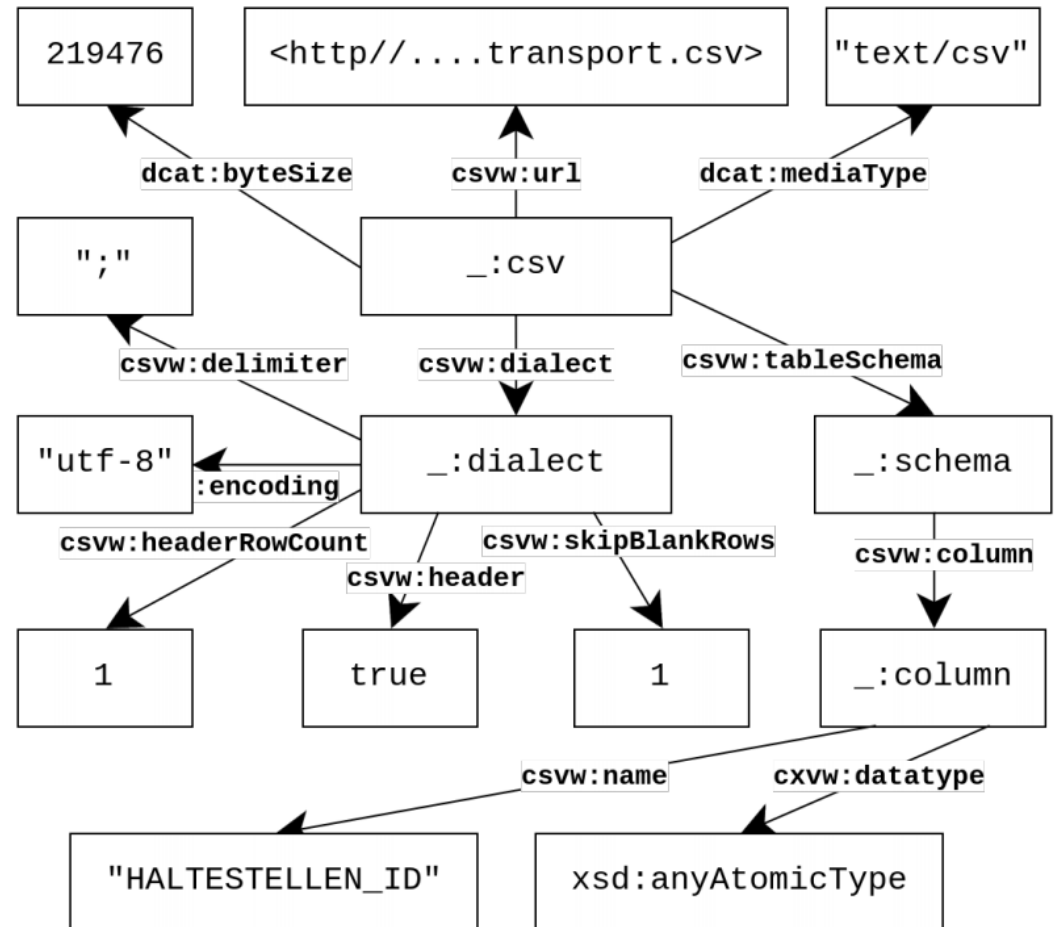
- Schema properties:

CSV header line

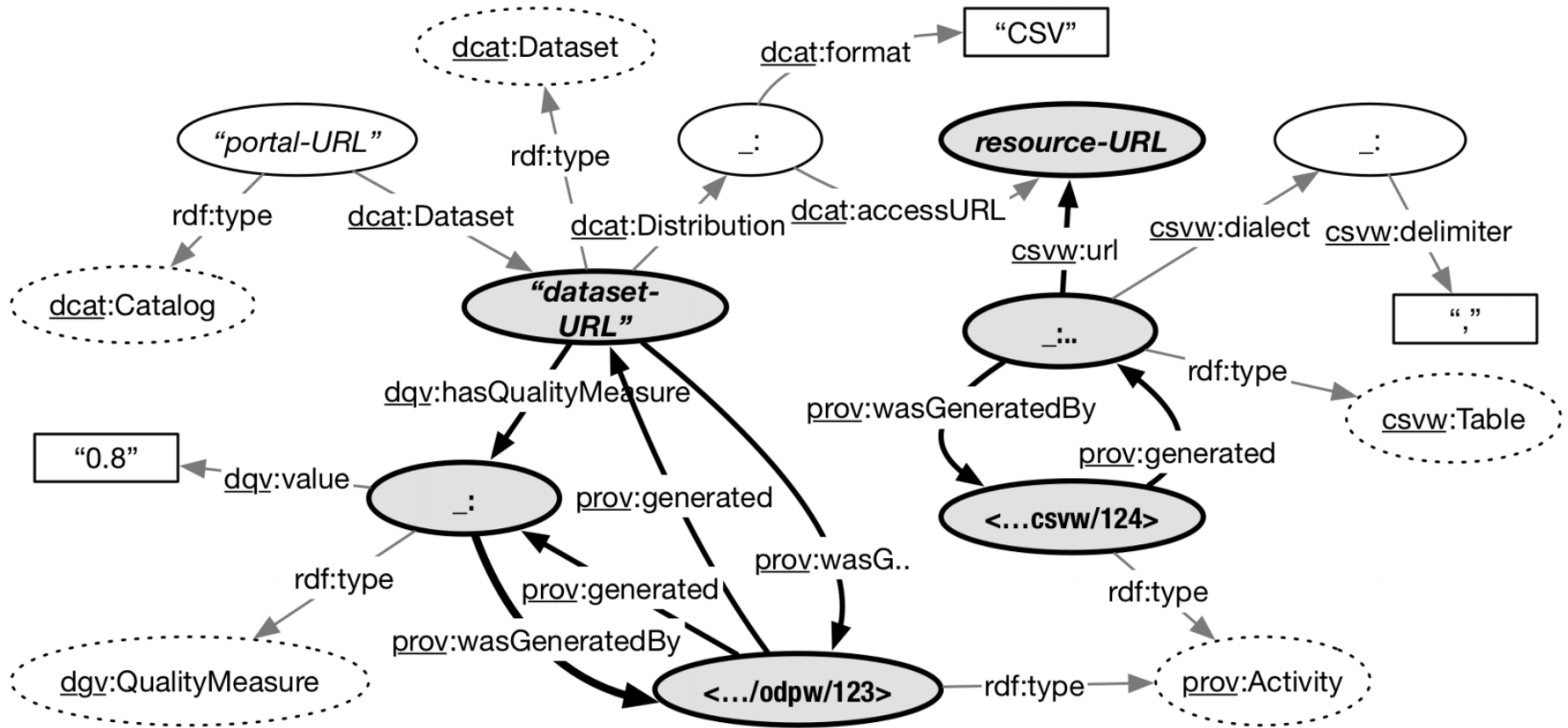
→ csvw:name

Column type detection

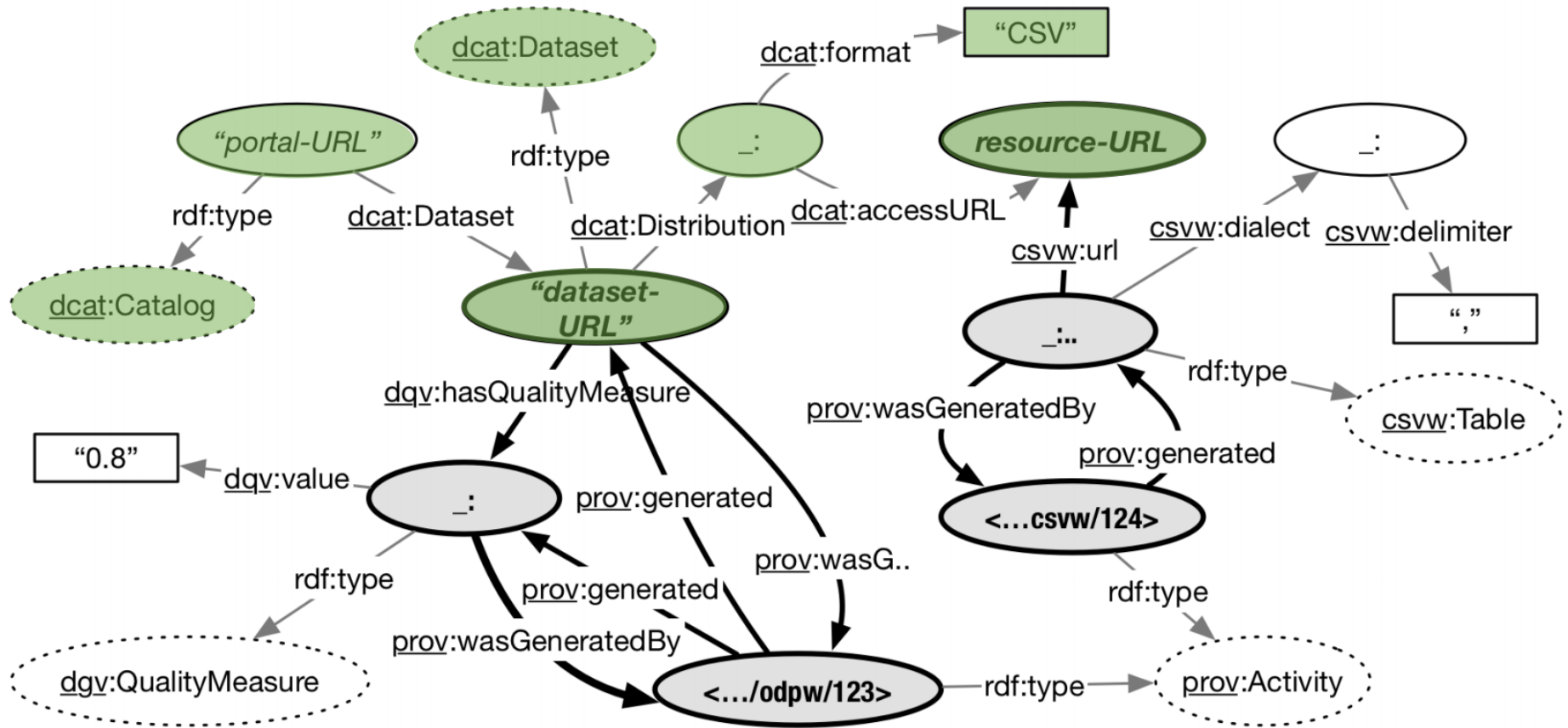
→ xsd:datatype



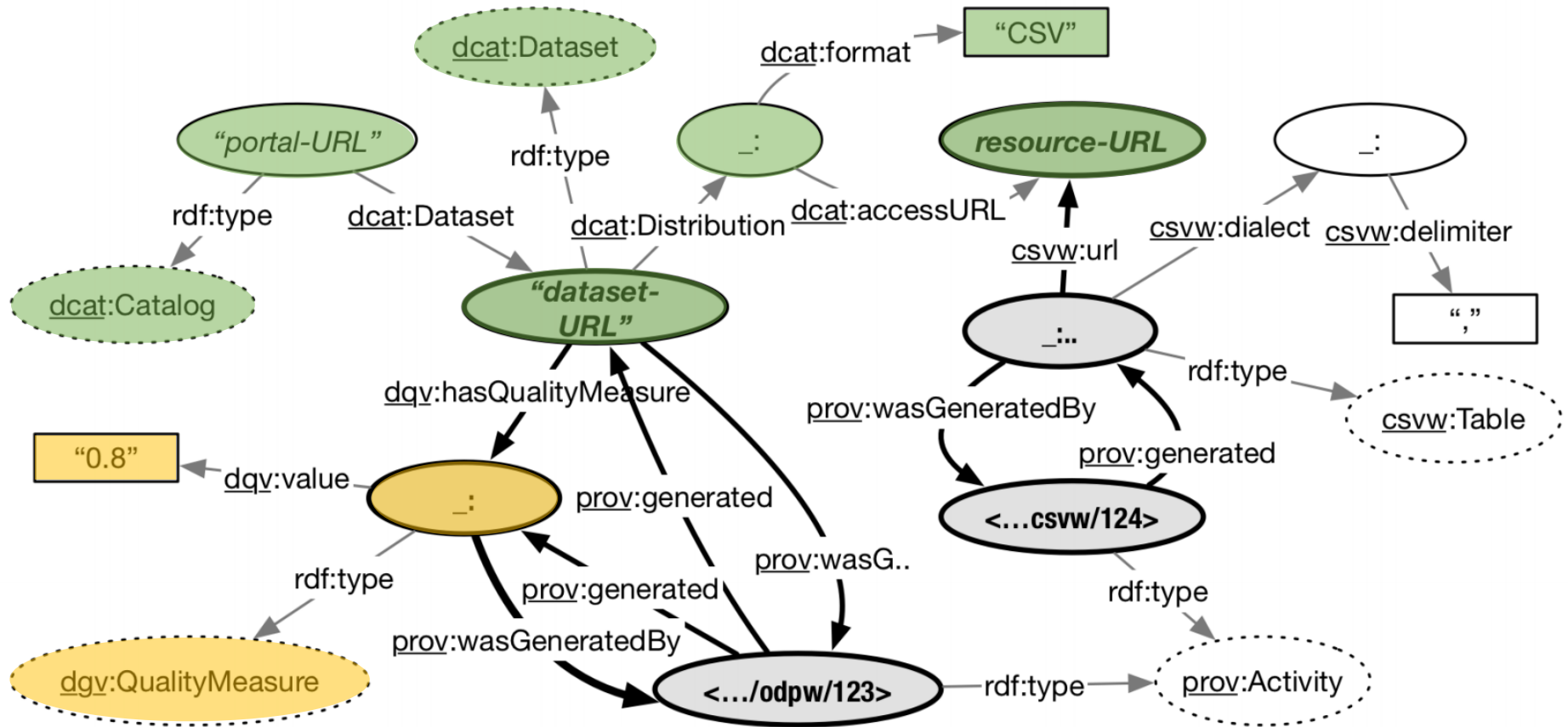
The big picture



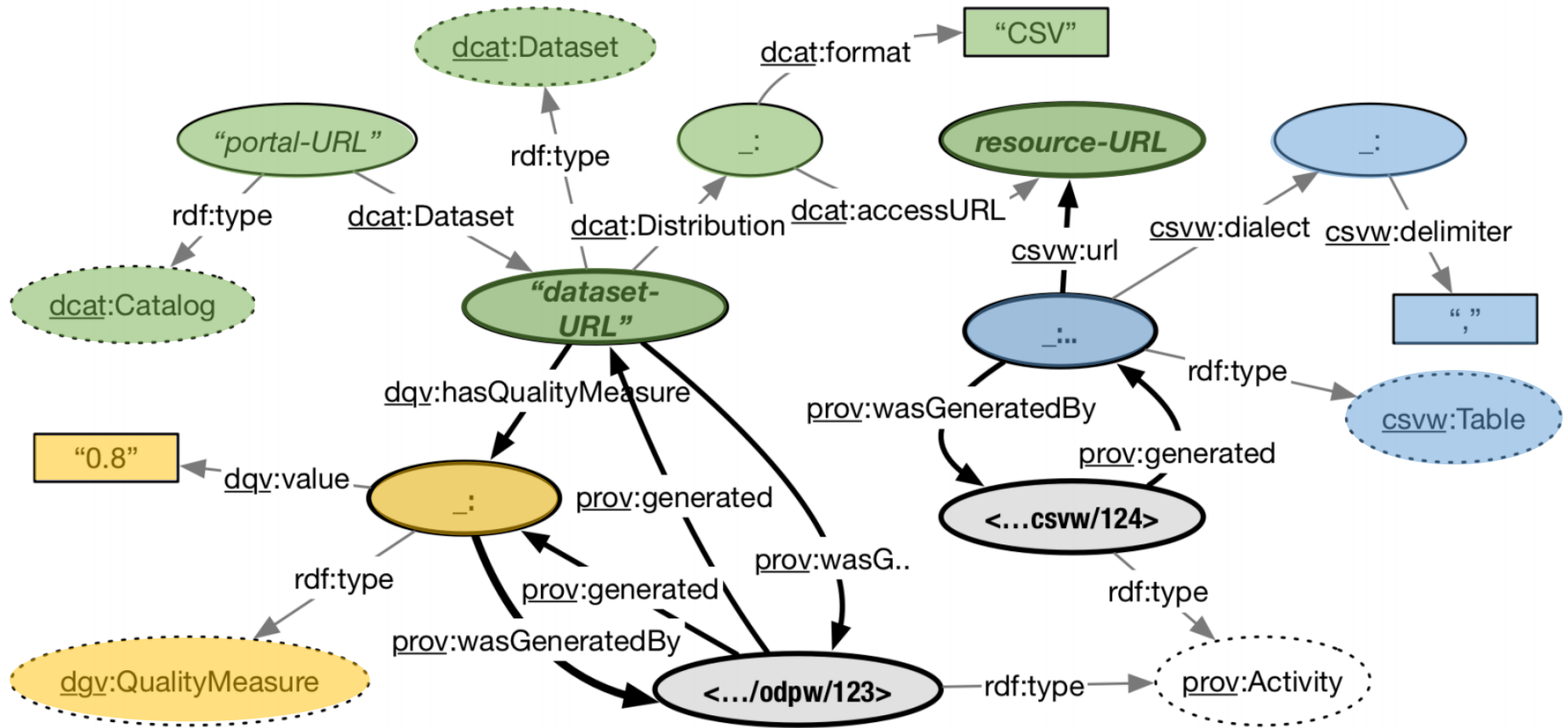
The big picture



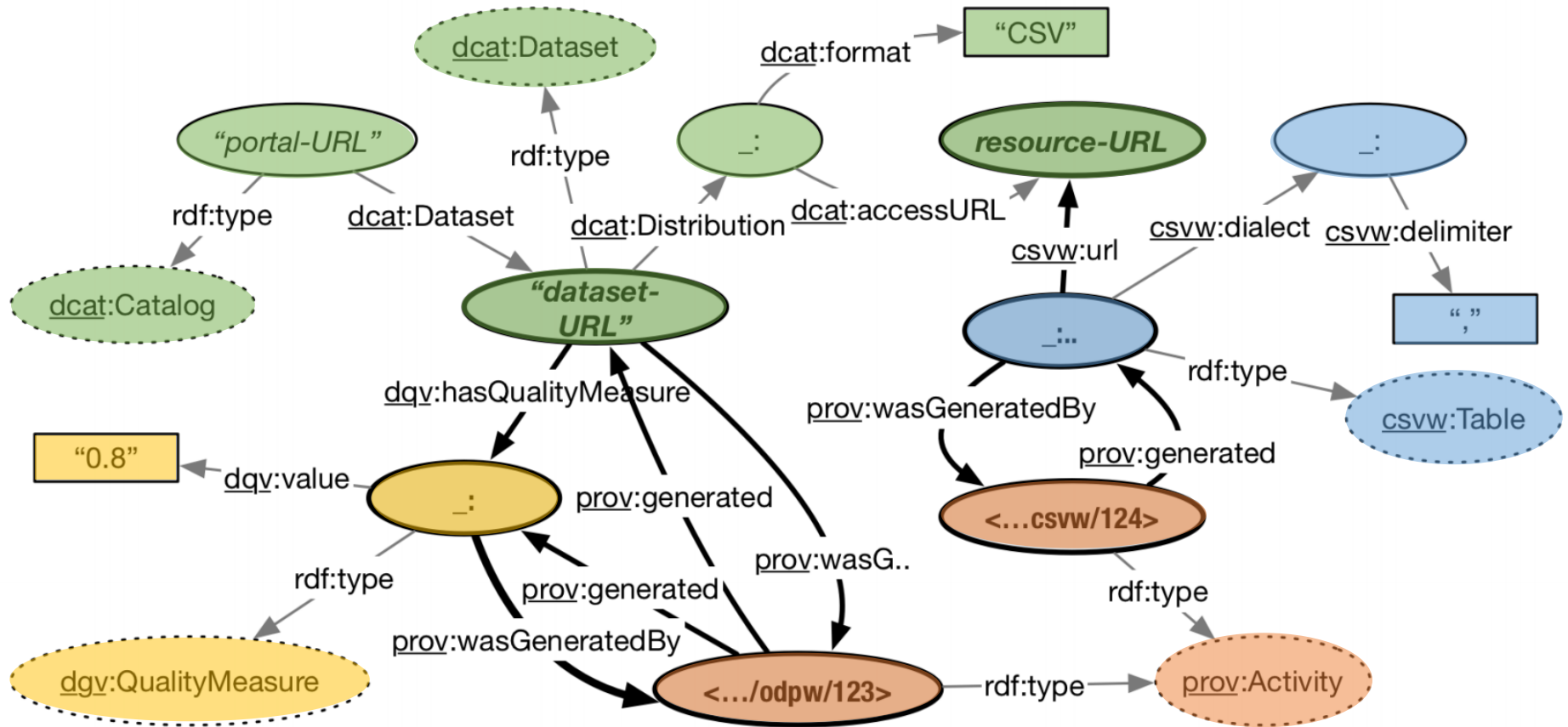
The big picture



The big picture



The big picture



Enable Access

- SPARQL endpoint [1]:
 - Three versions in RDF triple store (as named graphs)
 - 120 million triples each
- Memento framework:
 - Datetime negotiation on “Accept-Datetime” HTTP header
 - Access to original metadata, DCAT, and DQV measures
- Schema.org via sitemap.xml [2]:
 - Publishing of all 850k datasets as HTML-embedded Schema.org

[1] <http://data.wu.ac.at/portalwatch/sparql>

[2] <http://data.wu.ac.at/odso/>

Outlook & Future Work

data.wu.ac.at/porta1watch

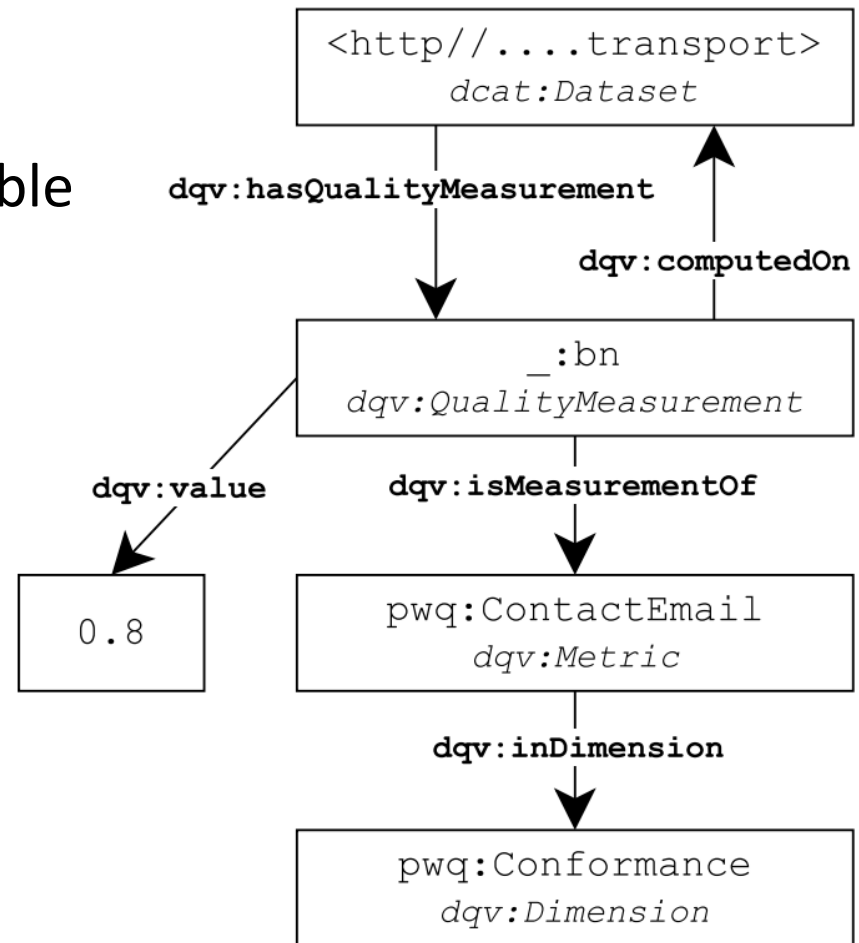
- Richer CSVW metadata
 - XSD datatypes, date(time) patterns, ...
- Access/query historical data
 - Named graphs vs change-based vs timestamp-based
- Add links to datasets and external knowledge
 - Enrich/link metadata (e.g., tags/keywords)
 - Derive semantic labels from data sources



Backup Slides

Portal Watch quality measures

- Existence
 - Is certain metadata available
- Conformance
 - Valid emails/URLs/dates
- Open Data
 - Open format
 - Machine-readable
 - Open license



Add provenance information

- We annotate:
 - Mapped DCAT description
 - Quality measurements
 - CSVW metadata
- PROV-Activities:
 - „fetch“-activity
 - „csvw“-activity

